5.6.2 Secondary Treatment

The secondary treatment includes treatment units such as filters, aeration tanks, activated sludge process, oxidation ponds , aeration lagoons, septic tanks, Imhoff tanks, etc.

1. Filters

The different types of filters that may be used in sewage treatment are

i. Contact beds (or) Contact filters

ii. Intermittent Sand filters

iii. Trickling filters

2. Activated Sludge Process

The effluent from primary sedimentation tank is mixed with activated sludge (20-30%) containing a large concentration of active aerobic micro organisms and with large quantity of air for about 4 to 8 hours.

Under these conditions, the micro-organisms will oxidise the organic matter, and the suspended and colloidal matter tend to coagulate and form a precipitate which settles down in the secondary settling tank. This settled sludge is known as activated sludge. The process of settling the activated sludge is known as activated sludge process.

5.6.3 Tertiary Treatment

Nitrogen and Phosphorous which are not removed in the secondary treatment are removed by tertiary treatment. This includes disinfection and removal of heavy metals.

1. Disinfection

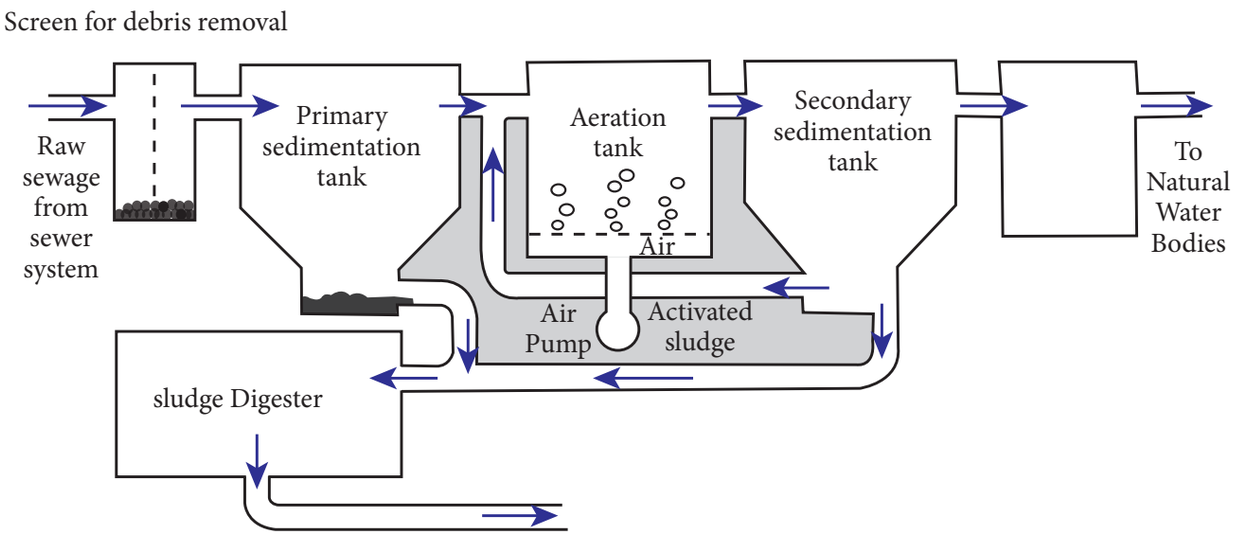
It is the process by which the pathogenic bacterias can be removed by adding chlorine with effluent.

2. Removal of heavy metals

The heavy metals like nitrogen and phosphorous due to industrial waste in the sewage can be removed by suitable methods.

Activity 2

Visit a sewage treatment plant in your town and submit a report.



Secondary Sewage Treatment

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